

Customer Segmentation Analysis Report

Introduction

Customer segmentation is an important data analytics technique used to classify customers into groups based on their characteristics and purchasing behavior. This project analyzes shopping mall customer data using Python and the K-Means clustering algorithm.

Objective

Analyze customer data, clean and preprocess the dataset, visualize customer characteristics, apply K-Means clustering, identify valuable customer groups, and provide business recommendations.

Dataset Description

The Mall Customers dataset contains 200 customer records with CustomerID, Gender, Age, Annual Income (k\$), and Spending Score (1-100).

Data Cleaning

The dataset was checked for missing values, duplicates, and incorrect data types. No missing values were found. Relevant numerical features were standardized using StandardScaler.

Exploratory Data Analysis

Histograms and scatter plots were created to analyze customer age, annual income, and spending behavior. NumPy was used to calculate summary statistics.

K-Means Clustering

The clustering model was built using Annual Income and Spending Score. Customers were grouped into five clusters based on similar characteristics.

Elbow Method

The Within-Cluster Sum of Squares (WCSS) was calculated for K values from 1 to 10. The elbow point at K = 5 indicated the optimal number of clusters.

Customer Segments

The analysis identified five groups: Moderate Customers, Low-Value Customers, High Income–Low Spending Customers, Low Income–High Spending Customers, and Premium Customers.

Business Insights

Premium customers should receive exclusive rewards. High-income customers with low spending can be targeted with personalized promotions. Loyalty programs can improve retention.

Recommendations

Implement VIP memberships, personalized marketing campaigns, bundled offers, customer loyalty programs, and periodic segmentation updates.

Conclusion

The project successfully applied K-Means clustering to identify meaningful customer segments and generate business insights that support data-driven marketing decisions.